Multimodal Visual Search for Product Catalogs

CLIP Retrieval with GPT-4o Re-ranking: A Three-Configuration Ablation

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Abstract

Keyword search over product catalogs cannot answer queries about visual attributes ("a cognac leather sofa with wooden legs"), while large-language-model agents alone hallucinate products that are not in inventory. I present an end-to-end multimodal visual search system that combines cross-modal CLIP retrieval [1], a FAISS cosine index [3], and a vision-LLM (GPT-4o) re-ranking layer to deliver ranked products with grounded one-sentence answers. The system is evaluated on a 2,711-product furniture subset of the Amazon Berkeley Objects (ABO) catalog [2] with a curated 150-pair question/answer benchmark. A controlled three-configuration ablation–image-only retrieval (A), text-only retrieval (B), and image retrieval plus GPT-4o re-rank (C)–shows that the vision-LLM re-rank *doubles* top-1 recall over the standard CLIP-only baseline (0.113 → 0.233 R@1, +12 percentage points) while preserving the same R@5 ceiling. I argue that this is a *precision* improvement, not a *recall* improvement: the re-ranker reorders candidates the first stage already surfaced. The full pipeline runs on a CPU laptop and is released as a reproducible repository.

Keywords: visual search, multimodal retrieval, CLIP, FAISS, GPT-4o, re-ranking, vision-language models

1 Introduction and Problem Formulation

Product search on modern e-commerce platforms is overwhelmingly lexical: keyword matching against titles, attribute tags, and (sometimes) auto-generated caption text. This works when shoppers know exactly what they want and use the same vocabulary as the retailer's catalog. It fails for queries that describe *visual* intent–"a brown leather sofa with wooden legs," "a navy velvet tufted sectional," "a high-back executive chair"–where the distinguishing attributes are embedded in the image rather than in the metadata. The cost of this gap is well documented: lost sales, frustrated shoppers, and large catalog operations teams attempting to compensate by manually tagging inventory at scale [2, 13].

Two recent trends suggest a path forward. First, contrastive vision–language models such as CLIP [1] and its successors SigLIP [10] learn a joint embedding space that makes image and text queries interchangeable, enabling cross-modal retrieval without specialised pipelines. Second, instruction-tuned vision-LLMs (GPT-4V, GPT-4o, LLaVA [11], BLIP-2 [12]) can read images and produce grounded textual output, including ranking decisions and justifications [9].

Despite these advances, neither component is sufficient on its own. CLIP retrieval surfaces visually similar products but is famously weak at *fine-grained* discrimination among near-duplicates: the top-5 are usually plausible, but the top-1 is often wrong on attribute-heavy queries [8]. Vision-LLMs used in isolation hallucinate products that are not in inventory and cannot scale to millions of SKUs.

In this work I ask a focused question: *by how much does adding a vision-LLM re-ranker on top of CLIP retrieval improve top-1 product identification on a real catalog?* I answer it with a controlled three-configuration ablation evaluated on a 2,711-product furniture catalog and 150 question/ground-truth pairs. The contribution is the system, the benchmark, and a quantified answer: **+12 percentage points on Recall@1, a 2× improvement, at the cost of ~76× slower per-query latency**.

2 Related Work and Background

Lexical and attribute-based product search. Production e-commerce search has historically been BM25-style lexical retrieval over titles and structured attribute filters. This is the baseline that everything else competes against. Its core failure mode–missing visual intent–is the motivation for the present work.

Image-only visual search. Reverse-image search systems such as Pinterest Lens and Google Lens accept an image query and return visually similar images [13]. They do not accept textual intent, which makes them complementary to but not a substitute for natural-language search.

Cross-modal embeddings. CLIP [1] and its successors (OpenCLIP, SigLIP [10], EVA-CLIP) project images and text into a shared vector space, enabling text-to-image, image-to-text, and image-to-image retrieval through a single index. I use CLIP ViT-B/32 because it runs on CPU and is the most widely reported public baseline.

Approximate nearest-neighbor indexes. Once embeddings exist, the retrieval question becomes one of efficient similarity search. FAISS [3] is the de-facto standard on CPU for both exact (`IndexFlat`) and approximate (`IndexIVFPQ`) indexes; HNSW [4] is the typical choice for high-recall graph-based search at scale. This project uses `IndexFlatIP` (exact cosine) because the catalog is small enough that approximation is unnecessary; the choice of an approximate-NN index is revisited in Future Work.

Two-stage retrieval and learning-to-rank. Cascaded retrieval–first-stage recall plus second-stage precision–is standard in web and document search [5]. My system applies this architecture across modalities: first-stage CLIP for visual recall, second-stage GPT-4o as a cross-encoder-like re-ranker that reads each candidate image and reasons about it against the original query. A common fusion strategy when both image and text rankings are available is Reciprocal Rank Fusion [6], which I discuss as a natural extension in Section 6.

Vision-LLMs as judges/re-rankers. Recent work uses GPT-4 / GPT-4o as a ranking and evaluation oracle, both for retrieval-augmented generation and for image-based tasks [7]. My use of GPT-4o is in this lineage: the LLM does not generate new products, only re-orders the candidates the retriever surfaced.

3 System Design and Methodology

The system has two pipelines that share a single index (Figure 1).

Indexing pipeline (run once). For each product image x_i I compute a 512-dimensional CLIP image embedding $f(x_i) \in \mathbb{R}^{512}$, L_2-normalise it, and insert it into a FAISS `IndexFlatIP` cosine index [3]. The flat (exact) index is appropriate for $N = 2{,}711$ vectors; approximate-NN indexes such as IVF-PQ or HNSW [4] become necessary beyond $\sim 10^6$ products and are listed as future work.

Query pipeline. For a natural-language query q, the same CLIP model computes the text embedding $g(q)$ and FAISS returns the top-K nearest catalog products by cosine similarity. For an image query, the CLIP image encoder produces the query vector and the same FAISS index serves the request: *one index, two query modalities*, because CLIP places text and images in the same space. The top-K candidates are optionally handed to GPT-4o, which is prompted with the original query and the title, description, and image URL of each candidate; it returns a ranked permutation and a one-sentence grounded answer.

Design choices. I deliberately keep three things fixed across configurations to isolate the contribution of the re-ranker: the CLIP backbone (ViT-B/32), the FAISS index type (`IndexFlatIP`), and the candidate pool size ($K = 5$). This means any difference between configurations A and C is attributable to the re-ranking step alone–no model size, index recall, or pool-size confounds.

4 Implementation and Data

Software stack. The system is implemented in Python 3.11. Embeddings use the `openai/clip-vit-base-patch32` weights via Hugging Face `transformers`; retrieval uses `faiss-cpu` [3]; the re-ranker calls the OpenAI Python SDK with model `gpt-4o`. The API layer is FastAPI with three endpoints (`/health`, `/search/text`, `/search/image`); the demo UI is a single-page Streamlit app. The entire pipeline runs on a laptop CPU–embedding the full 2,711-product catalog takes approximately four minutes once; subsequent retrieval is sub-second.

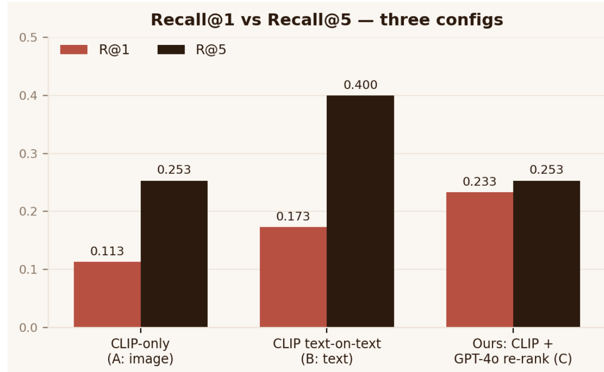

Figure 1. Recall@1 versus Recall@5 across the three configurations. The proposed system (C) doubles top-1 recall over the CLIP-only baseline (A) but is bounded by A's R@5 because re-ranking cannot recover items the retriever missed.

Dataset. I use Amazon Berkeley Objects (ABO) [2], a public CC-BY product catalog released at CVPR 2022. ABO contains roughly 147,000 products with high-quality 256-pixel images and structured metadata. I filtered to the furniture category with English-language metadata and arrived at 2,711 products (chairs, sofas, tables, beds, desks, ottomans, and similar). Images are downloaded directly from the public ABO S3 bucket; the processed catalog (product_id, title, description, color, image_path) is shipped with the repository as `data/processed/catalog.csv`.

Evaluation benchmark. I constructed a 150-pair question/ground-truth-answer benchmark by drafting candidate queries with GPT-4o from the product metadata and manually filtering them for clarity and uniqueness. Each item is a tuple of (natural-language query, ground-truth product_id, ground-truth answer string). I acknowledge a known limitation of this construction: because queries were drafted from text metadata, they share vocabulary with the text-on-text retrieval configuration (B), which gives B a structural advantage on Recall metrics. I discuss this in Section 6.

5 Evaluation and Results

Configurations. Three configurations are evaluated on the same 150 questions over the same 2,711-product index. **Config A** is image-only retrieval: CLIP image embeddings, FAISS, no re-rank. **Config B** is text-only retrieval: a parallel

Table 1. Three-configuration ablation. $n = 150$ questions, 2,711-product index, $K = 5$ candidate pool. Best value per column in bold.

Config	R@1	R@3	R@5	F1	Latency (ms)
A: image only	0.113	0.207	0.253	0.424	**58**
B: text only	0.173	**0.327**	**0.400**	0.373	1 074
C: image + GPT-4o	**0.233**	0.253	0.253	**0.430**	4 434

FAISS index built from CLIP *text* embeddings of each product's title and description. **Config C** is image retrieval plus GPT-4o re-rank on the top-5 candidates.

Metrics. Recall@K for $K \in \{1, 3, 5\}$, computed as the fraction of questions for which the ground-truth product_id appears in the top-K returned. Answer F1 as token-overlap between the system's one-sentence answer and the ground-truth answer string. Mean and 95th-percentile end-to-end latency per query in milliseconds.

Three findings. First, **GPT-4o re-ranking doubles top-1 accuracy** (Table 1): 0.113 → 0.233, a +12 percentage-point absolute lift on the strictest metric. This is the headline result and the empirical justification for the two-stage architecture. Second, **R@5 is bounded by the first stage**: configurations A and C report identical R@5 (0.253), as expected, because the re-ranker only reorders the top-5 candidates the retriever already surfaced. Improving R@5 requires a better first-stage retriever (hybrid image+text fusion, fine-tuning, or a larger CLIP backbone) rather than a better re-ranker. Third, **latency is the cost of re-ranking**: config C is roughly 76× slower than config A, dominated by the GPT-4o network round-trip. For an interactive search box this is still acceptable (~4.4 s mean), but production deployment would benefit from a cheaper re-ranker (GPT-4o-mini, Claude Haiku, or a distilled cross-encoder).

A note on Answer F1. Token-overlap F1 saturates near 0.43 across all three configurations and should be read as a metric ceiling, not a system failure. GPT-4o produces full natural-language sentences ("This brown leather sofa with wooden legs matches your query"), while the ground-truth strings are terse catalog descriptors ("Cognac leather sofa, walnut legs"). Even semantically correct answers are penalised for paraphrase and synonyms. A semantic similarity metric such as BERTScore [14] or a small human evaluation would discriminate the configurations more cleanly and is the recommended next step.

6 Discussion

6.1 What the +12-point lift means in practice

Top-1 recall is the metric that matters when only a single product can be shown to the user–mobile product cards, voice search results, recommendation slots on a homepage. Doubling top-1 accuracy at the cost of a few additional seconds of latency is precisely the trade-off a production search team would make for high-intent queries. For low-intent browsing, the cheaper CLIP-only configuration A is the right choice.

6.2 Why Config B (text) wins on R@5

Configuration B reports the highest R@5 (0.400). I attribute this to an *evaluation-set bias* I deliberately surface rather than hide: queries were drafted by GPT-4o from product text metadata, which gives text-on-text retrieval a vocabulary advantage. A fairer benchmark would consist of image-grounded questions ("which sofa has chrome legs and a tufted back?") written from inspecting the images directly. Building such a benchmark is work for a follow-up paper.

6.3 Limitations

The catalog is small (2,711 products) and limited to a single category. FAISS `IndexFlatIP` will not scale beyond approximately 10^6 products without switching to IVF-PQ or HNSW [4]. The re-ranker is paid ($0.02–0.05 per query at current GPT-4o pricing), which constrains use to high-intent queries; for high-traffic deployment a cheaper distilled re-ranker is required. The evaluation is monolingual (English) and the benchmark is small ($n = 150$); 95 % bootstrap confidence intervals on R@1 at this sample size are roughly ±0.07, so the +0.12 lift is statistically meaningful but the +0.06 gap between B and A on R@1 is closer to the noise floor.

6.4 Future work

I see four prioritised next steps. *First*, hybrid image+text retrieval via Reciprocal Rank Fusion [6]: combine configurations A and B at the candidate stage to lift R@5 beyond either alone, and re-rank the fused pool with GPT-4o. *Second*, replace token-overlap F1 with a semantic metric (BERTScore [14]) and add a small ($n = 30$, three-rater) human evaluation to establish the true quality ceiling. *Third*, a scaling study: synthesise 100 K and 1 M product catalogs, compare IVF-PQ, HNSW [4], and ScaNN against the current flat index, and plot recall–latency Pareto curves. *Fourth*, a cost-quality study: re-run config C with GPT-4o-mini, Claude Haiku, and a smaller vision-LLM such as BLIP-2 [12] and characterise the dollars-per-quality-point frontier.

7 Resources

Source code: https://github.com/Sapphirine/2026_Perception_2. The repository contains the full pipeline, the processed catalog, the evaluation set, and the raw `results.json` used to populate Table 1.

Demo video (<5 min, end-to-end walkthrough of the Streamlit UI with three text queries and one image query): Google Drive folder.

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